

Individual Challenge

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Introduction

Based on the data foundation established in Phase 1 and the main research objective:

“What influences PM2.5?” I chose to work on Individual Challenge 1: Add a new question/problem that requires at least one additional data source.

For this challenge, I proposed a new question:

“Is a country’s average PM2.5 level related to its per capita GDP?”

To explore this, I introduced a new dataset titled “API_NY.GDP.PCAP.CD_DS2_en_csv_v”, which was obtained from the World Bank. This dataset contains annual per capita GDP data for countries around the world from 1960 to 2023.

I compared this new dataset with the average PM2.5 levels for each country and attempted to draw conclusions regarding potential relationships.

Integrated schema

Based on the integration schema and table structure previously completed, I added a new table named “GDP” to support the integration and linking of data (see Attachment 1).

Before the integration, I conducted data cleaning and transformation, such as:

Removing columns that were not relevant to this analysis (e.g., “Country Code”);

Renaming and harmonizing the fields to match the integrated schema.

Describing the attributes of the new table (see Attachment 2).

Analyze the data

First, based on the provided data, I created data marts using SQLite software. The data marts include three main tables: Year, GDP, and PM2.5-

Country.

a. Detail what analysis the data marts support:

1. Is a country’s average PM2.5 level related to its per capita GDP?

2. Analyzing how PM2.5 and GDP have evolved over time within individual countries

For the first question, we conducted a cross-sectional analysis by selecting ten representative countries with complete data.

We compared their per capita GDP and PM2.5 levels in 2020 and 2021 to highlight differences in development and environmental conditions.

For the second question, we used time-series line charts to visualize GDP and PM2.5 trends from 1998 to 2022 in selected countries.

This helps assess whether economic growth aligns with environmental improvement, indicating progress toward green development.

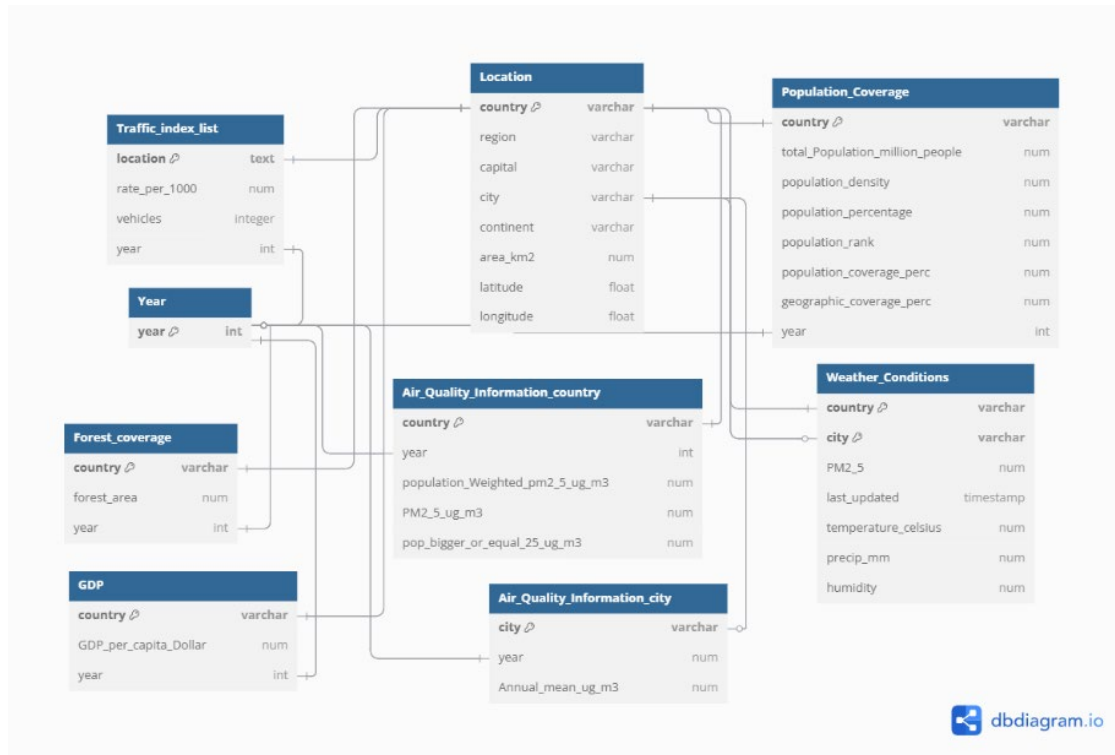
Result

To address the main question, I broke it down into two sub-questions and attached the relevant visualizations in Attachment 3 and Attachment 4.

From Attachment 3, our visual analysis shows that countries with higher GDP per capita generally have lower PM2.5 levels, while countries with lower incomes tend to experience higher air pollution, suggesting an inverse relationship between income and pollution.

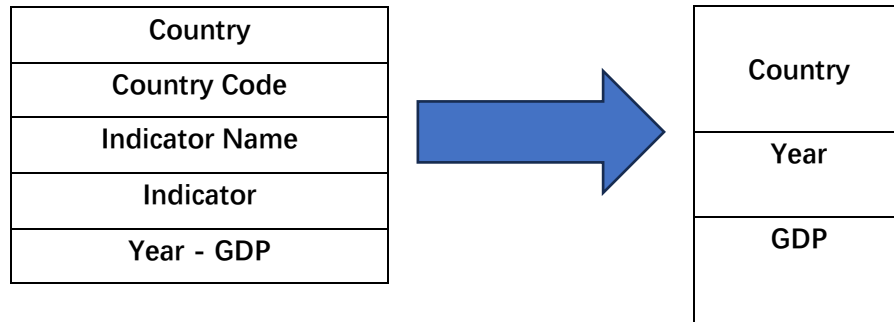
From Attachment 4, we observe that in developed countries (such as Poland, Italy, and South Korea), higher GDP per capita is associated with lower PM2.5. In contrast, in some developing countries (such as Mexico, Angola, and Chile), both GDP and PM2.5 rise together. China is a special case, as it transitions from a developing to a more developed profile after 2020.

Attachment 1



Attachment 2

- Removing columns which are useless



-Description of the data

Field	Description	Type of data	Example
Country	Name of the country	text	Portugal
Year	Year of the data	int	2020
GDP_dollar	GDP per capita	Numeric	2472.95

- Integrated data

Origin Dataset	From	Target	Type Corresp.	Description
GDP	GDP_Country _Name	Location_Cou ntry	1:1	Name of the country

Attachment 3

-2020

PM2.5 - 2020

Year ● 2020



GDP 2020

Year ● 2020



2021

PM2.5 - 2020

Year ● 2021



GDP 2020

Year ● 2021

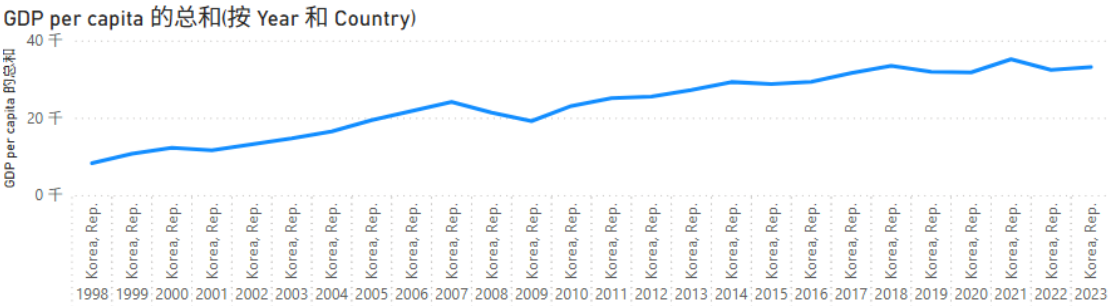
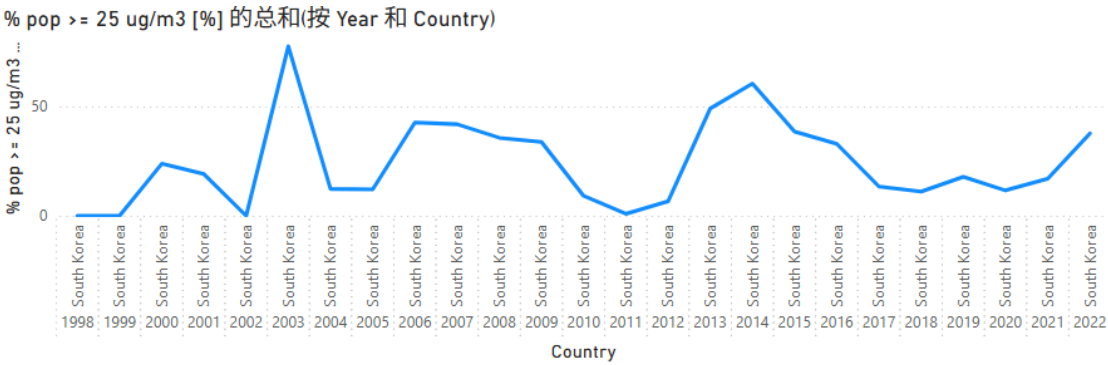


Attachment 4

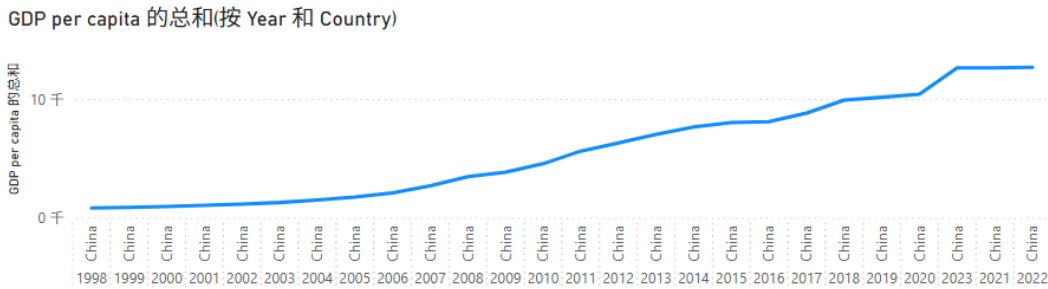
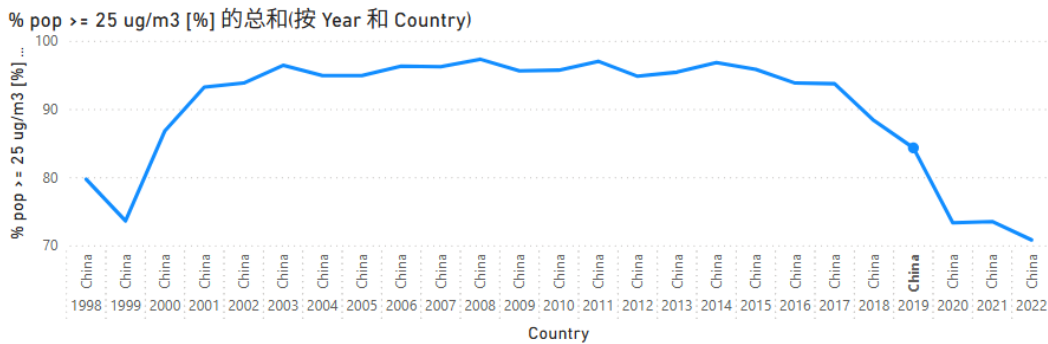
Page1-PM2.5

Page2-GDP

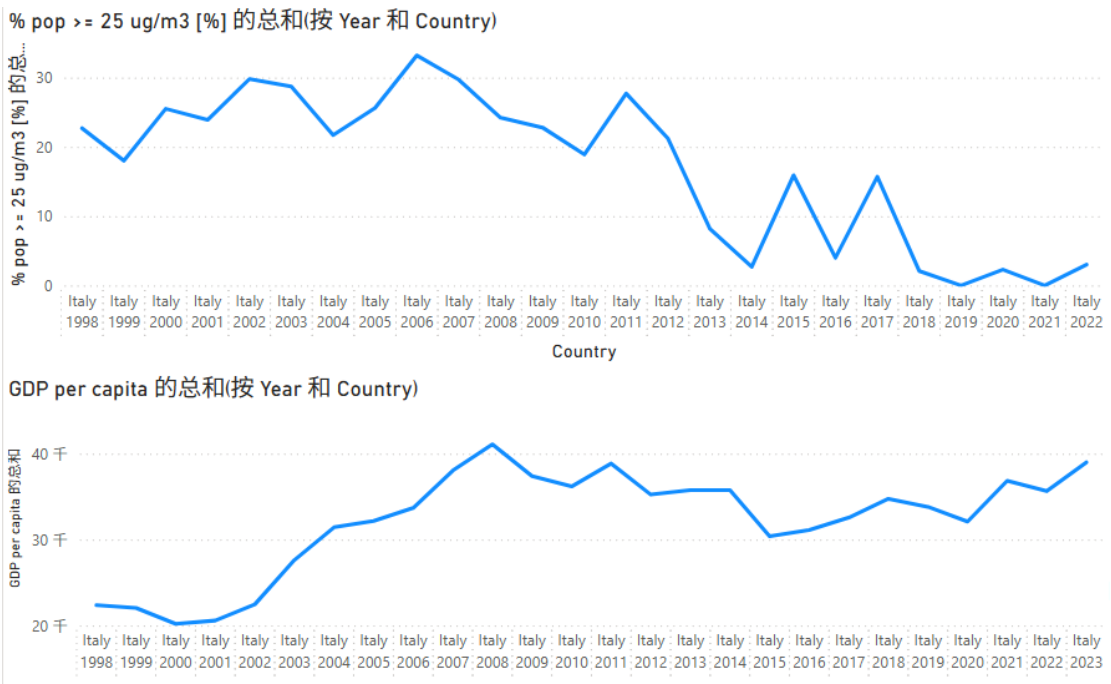
Soul Koria



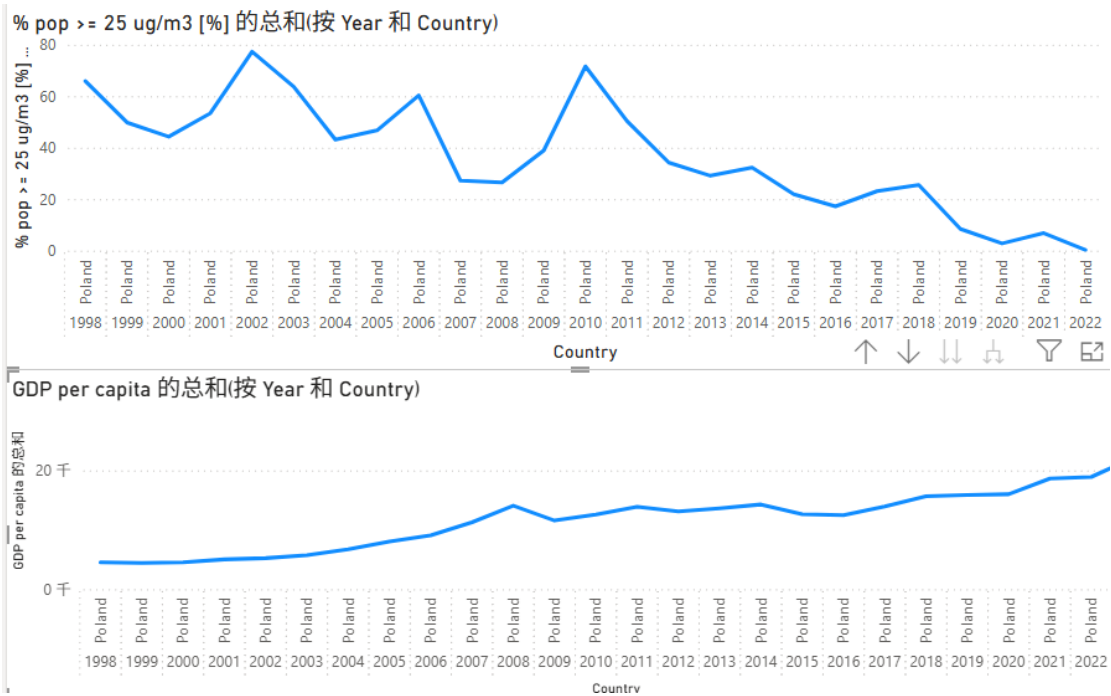
China



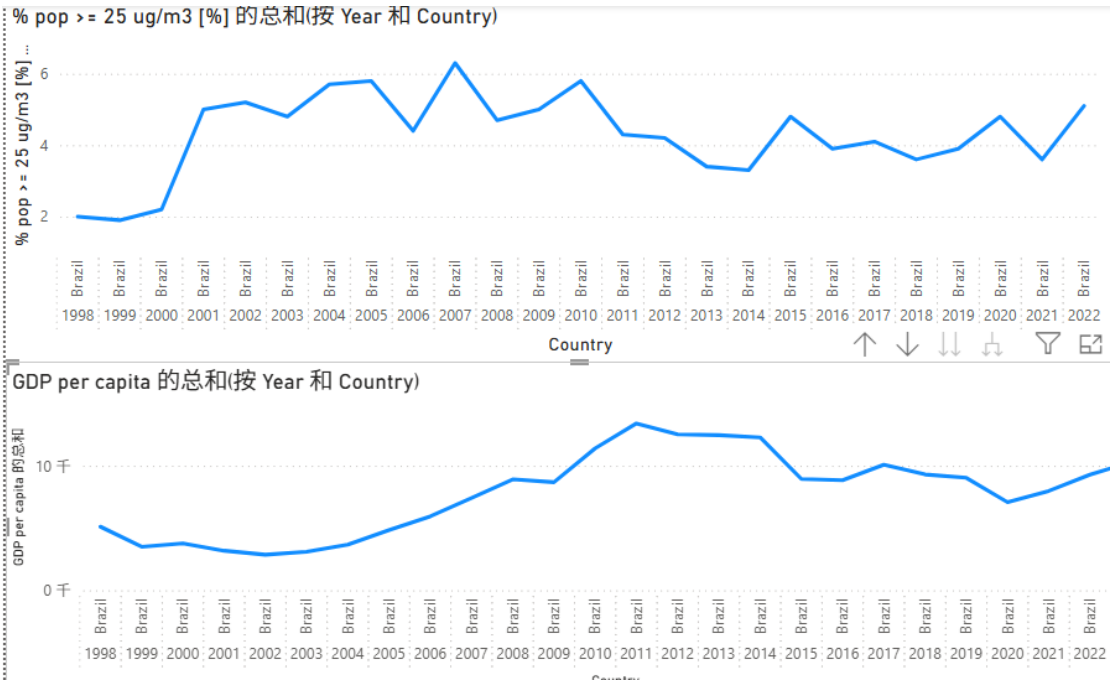
Italy



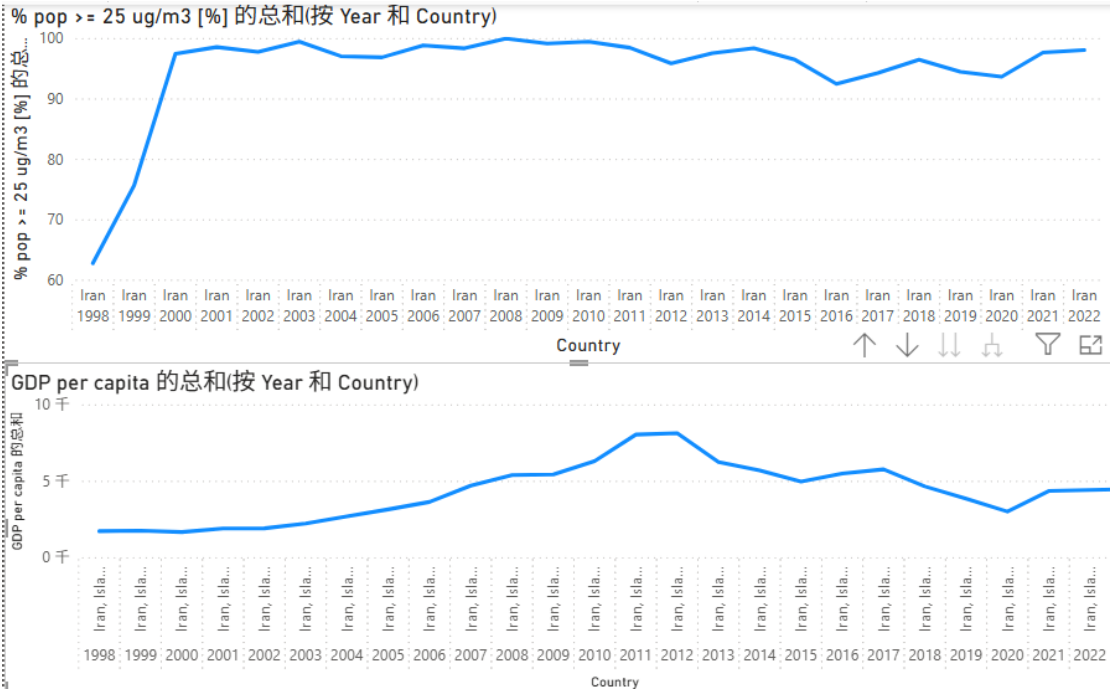
Polan



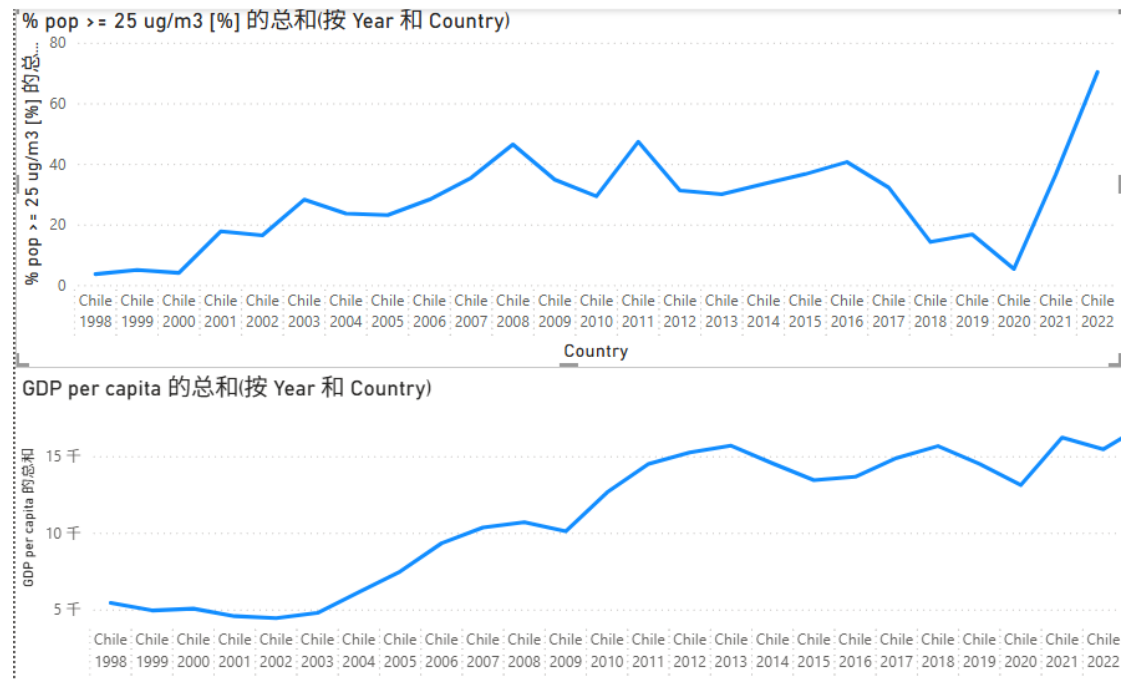
Brazil



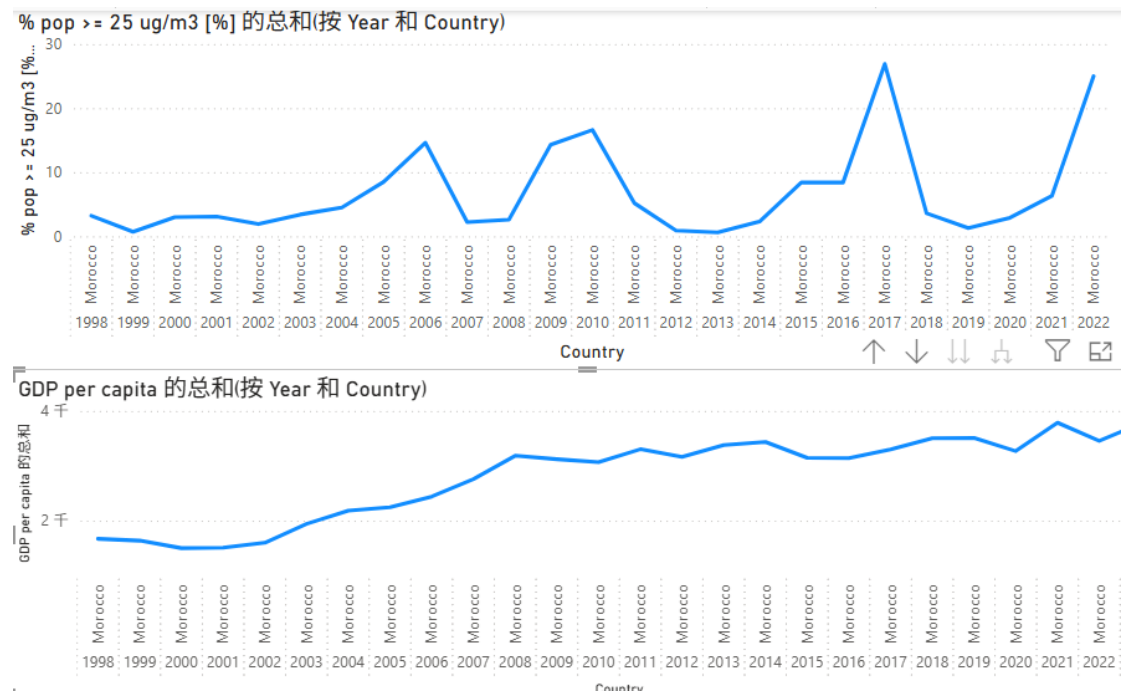
Iran



Chile

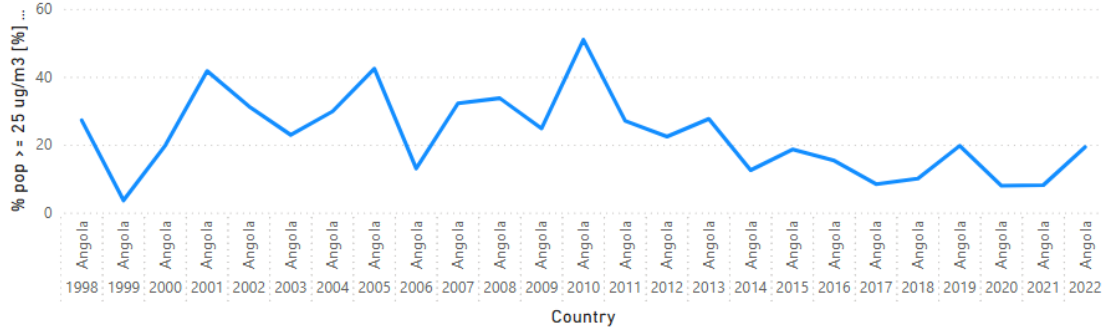


Morocco

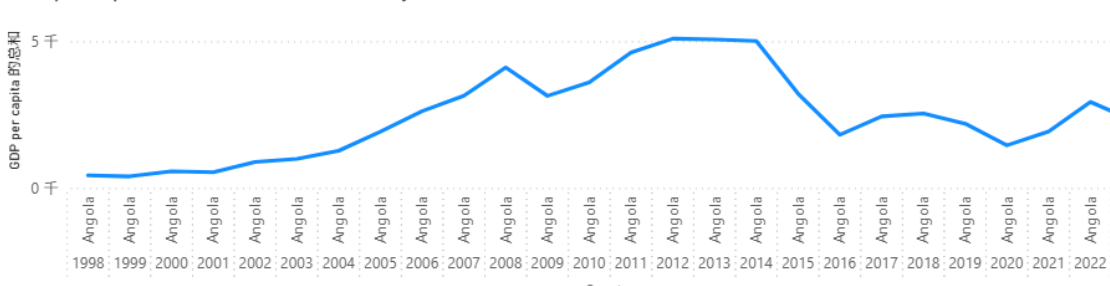


Angola

% pop >= 25 ug/m3 [%] 的总和(按 Year 和 Country)

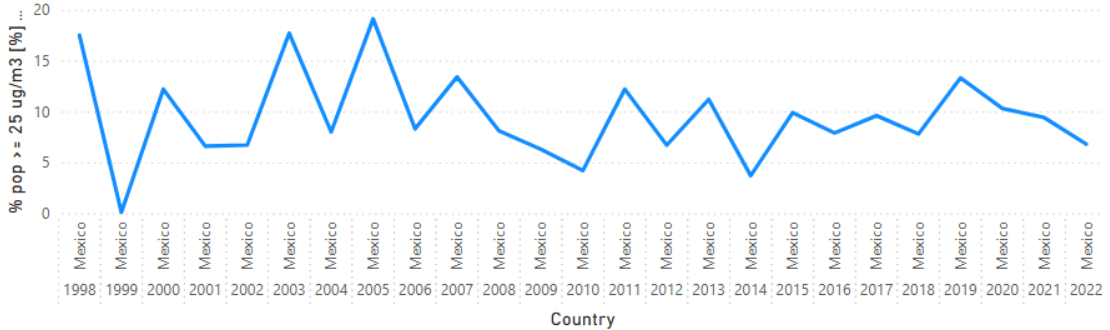


GDP per capita 的总和(按 Year 和 Country)



Mexico

% pop >= 25 ug/m3 [%] 的总和(按 Year 和 Country)



GDP per capita 的总和(按 Year 和 Country)

